

ENBC332: Homework #9

Due: Monday, 12/01/25 by noon, beginning of the class

Submit your homework to ELMS, including R code and dataset

Overall points: 40

Note: Late homework may be submitted up to 24 hours after the due date, but a 50% penalty will be applied.

Note: Please aim to write clean, well-organized code that is easy for others to understand. Be sure to include comments or explanations where needed. Points may be deducted if this is not done.

Note: Please name your codes using the following format: “problem1_fullname.R”. Start with the problem number and followed by your full name, e.g., “**problem1_jimmyazarnoosh**”.

Note: Using generative AI for learning purposes is allowed. However, directly copying or closely replicating AI-generated content is strictly prohibited and will result in a grade of **ZERO**. Repeated violations may lead to more severe consequences.

Note: Submit **a single zip file** containing the following:

- The dataset you used
- R code files
- Your answers to the questions in PDF format

Note: Functions are recommended to follow the format below:

e.g., calculating range.

```
# -----  
Data <- trees$Height  
my_range_func <- function(Data){  
  Range <- max(Data) - min(Data)  
  return(Range)  
}  
# -----
```

Dummy Variables

Problem 1. A hospital is studying the effect of a treatment (x_1) and patient gender (x_2) on recovery time (y) in days. The treatment is categorized as “Placebo” and “Drug”, while gender is categorized as “Male” and “Female”. (20 point)

The dataset is provided in the following:

Recovery Time	Treatment	Gender
10	Placebo	Male
8	Drug	Female
15	Placebo	Female
12	Drug	Male
7	Drug	Male
14	Placebo	Female
9	Drug	Female
11	Placebo	Male
6	Drug	Female
13	Placebo	Male

```
# Given dataset
Recovery_Time = c(10, 8, 15, 12, 7, 14, 9, 11, 6, 13)
Treatment = c("Placebo", "Drug", "Placebo", "Drug", "Drug", "Placebo", "Drug",
"Placebo", "Drug", "Placebo")
Gender = c("Male", "Female", "Female", "Male", "Male", "Female", "Female", "Male",
"Female", "Male")
```

Do the following tasks:

a) **Fit the Model:** Fit a linear regression model to predict recovery time based on treatment and gender and write the equation of the model. (10 point)

Note: Set “Female” and “Placebo” as the references.

b) Explain what the coefficients for the treatment and gender dummy variables represent. (5 point)

c) Predict the recovery time for: (5 point)

- a female patient who received the drug.
- a male patient who received the drug.

Hint: Use 0 or 1 in the regression model to calculate recovery time.

Multiple Linear Regression

Problem 2. A researcher is studying the factors affecting house prices in a city. They have collected data on 50 houses with the following variables: (20 points)

Price: The selling price of the house (in thousands of dollars).

Size: The size of the house (in square feet).

Bedrooms: The number of bedrooms in the house.

Age: The age of the house (in years).

Distance_to_City_Center: The distance of the house from the city center (in miles).

The dataset is provided in the following:

```
# Given dataset
set.seed(123) # For reproducibility
Price = round(rnorm(50, mean = 300, sd = 50), 2)
Size = round(runif(50, 800, 4000), 2)
Bedrooms = sample(2:6, 50, replace = TRUE)
Age = sample(1:50, 50, replace = TRUE)
Distance_to_City_Center = round(runif(50, 1, 30), 2)
```

Do the following tasks:

a) **Fit the Model:** Fit a linear regression model to the dataset and write the equation of the regression model based on the output of `summary(model)`. (5 points)

b) Which variables significantly impact the house price? Justify your answer based on the p-values. Consider the common significance level of 5%. (10 points)

c) **Prediction:** Predict the price of a house with the following attributes: (5 points)

- Size: 2500 square feet
- Bedrooms: 4
- Age: 20 years
- Distance_to_City_Center: 10 miles